Online Appendix

Delivering Degrees: Educational Spillovers from Amazon's Logistics Expansion

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This Online Appendix accompanies "Delivering Degrees: Educational Spillovers from Amazon's Logistics Expansion." It reports additional descriptive evidence, heterogeneity analyses, mechanism evidence, robustness checks, balance tests, and technical estimator details referenced in the main text but omitted there to satisfy the journal's page limit.

Contents

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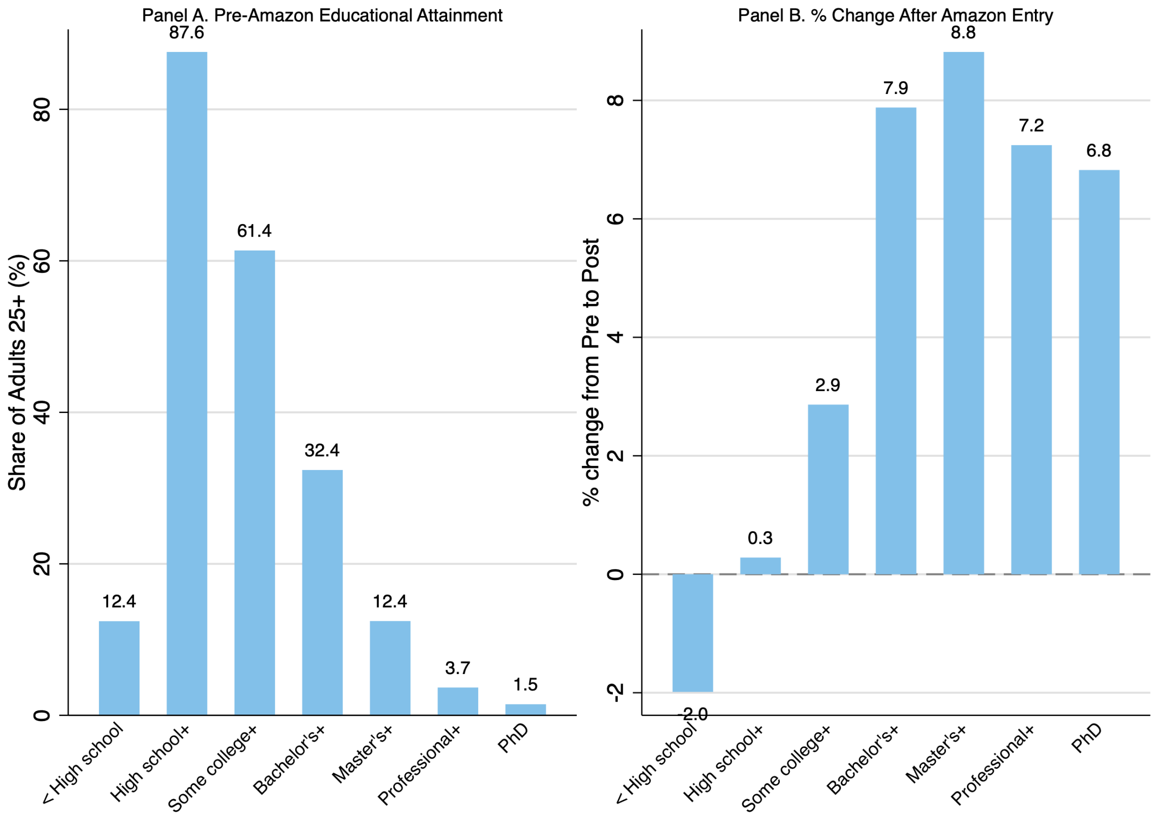
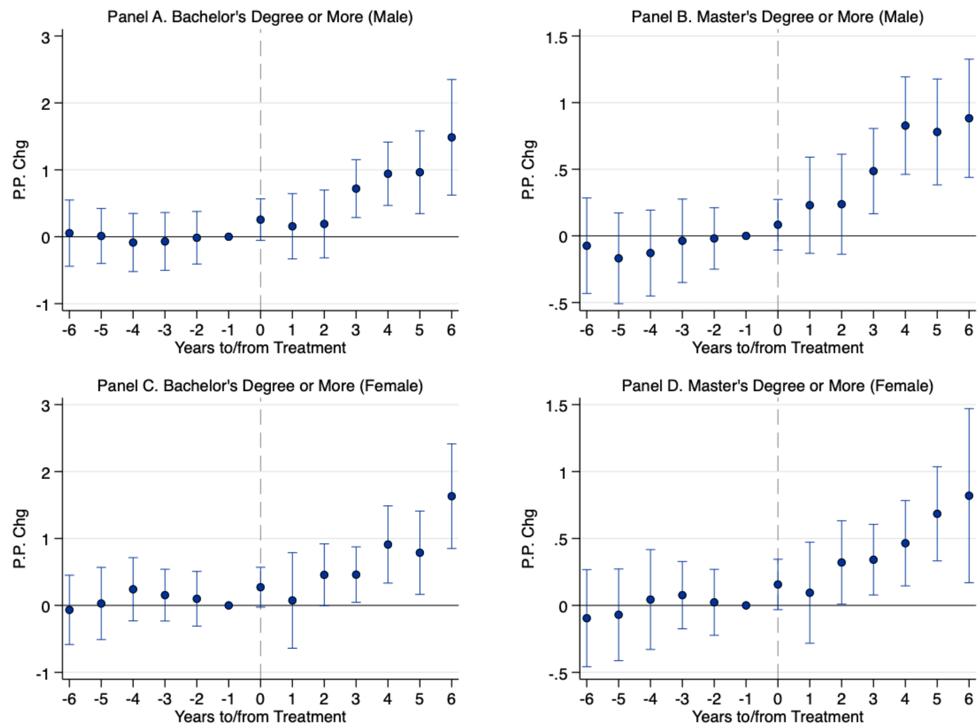
1 Additional Attainment Evidence

Figure A1: Educational Attainment Before and After Amazon FC Entry



Notes: This figure illustrates changes in educational attainment before and after Amazon fulfillment center entry within commuting zones. Panel A displays the pre-treatment distribution of educational attainment, measured as the percentage-point share of adults aged 25 and older who have completed at least the specified level of education: less than high school, high school diploma or equivalent, some college education, bachelor's degree, or master's degree or higher. Panel B presents the percentage change in each education category between the pre-treatment and post-treatment periods. The pre-treatment period is defined as the five-year average prior to Amazon's first fulfillment center opening in each commuting zone ($t = -5$ to $t = -1$), while the post-treatment period represents the five-year average following opening ($t = 0$ to $t = +5$). Data are from the American Community Survey (ACS) 1-Year estimates for 2010–2023, constructed from ACS microdata and aggregated to the commuting zone level using person weights. The sample consists of 170 commuting zones that eventually receive Amazon facilities during the study period.

Figure A2: Event-Study Estimates: Heterogeneous Effects by Gender on Educational Attainment



Notes: This figure presents dynamic estimates of Amazon fulfillment center entry on educational attainment, disaggregated by gender. The dependent variables measure percentage-point changes in the share of adults aged 25 and older who have completed at least a bachelor's degree (Panels A and C) or master's degree (Panels B and D), estimated separately for men and women. Data are from the American Community Survey (ACS) 1-Year estimates for 2010–2023, constructed from ACS microdata and aggregated to the commuting zone level using person weights. The vertical dashed line indicates the year of Amazon's first fulfillment center opening in each commuting zone, with year $t = -1$ serving as the omitted reference period. Estimates are obtained from a staggered difference-in-differences specification following Callaway and Sant'Anna (2021), which compares treated commuting zones to not-yet-treated control units while incorporating commuting zone fixed effects and year fixed effects. Each point represents an event-time average treatment effect on the treated (ATT), with vertical bars indicating 95% confidence intervals constructed using multiplier bootstrap standard errors clustered at the commuting zone level (999 replications).

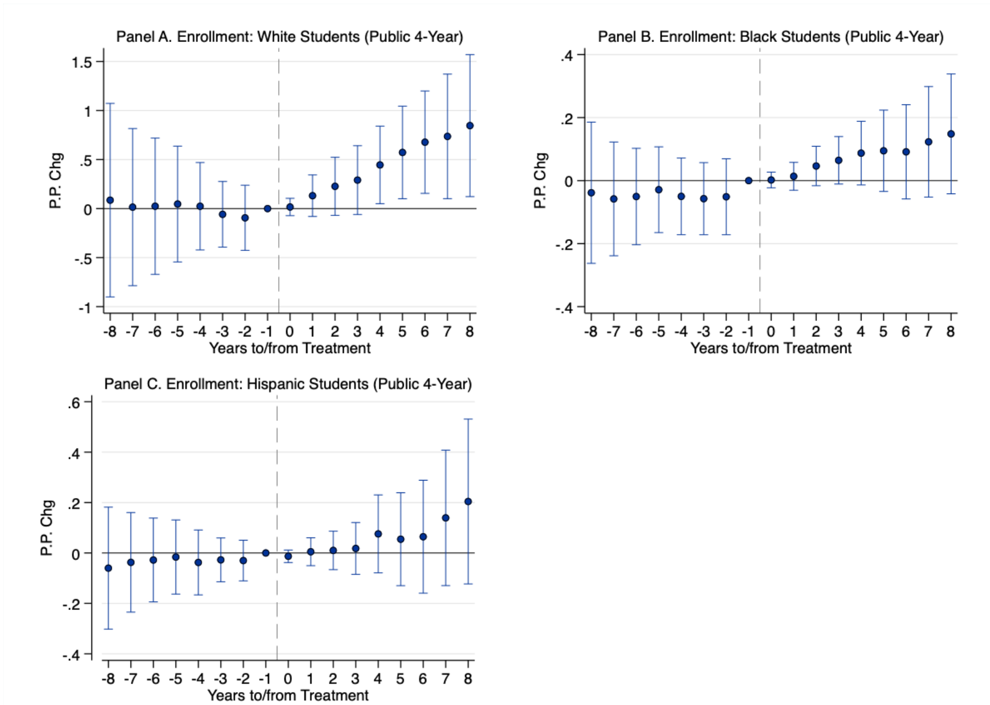
Table A1: Effects of Amazon's Entry on Educational Attainment by Gender

	High School+		Some College+		Bachelor's+		Master's+	
	Men	Women	Men	Women	Men	Women	Men	Women
Amazon Entry	0.055	0.357	0.143	0.595	0.580***	0.553***	0.454***	0.357***
	(0.148)	(0.222)	(0.218)	(0.400)	(0.181)	(0.195)	(0.132)	(0.133)
Mean Dep. Var.	88.43	88.43	59.87	59.87	28.88	28.88	10.84	10.84
SD	4.27	4.27	6.85	6.85	6.48	6.48	3.35	3.35
N	2,097	2,097	2,097	2,097	2,097	2,097	2,097	2,097

Notes: This table presents average treatment effects on the treated (ATT) for the impact of Amazon fulfillment center entry on educational attainment, disaggregated by gender. Data are from the American Community Survey (ACS) 1-Year estimates for 2010–2023, aggregated to the commuting zone level using person weights. The estimation sample consists of a balanced panel of 2,097 commuting zone-year observations. Estimates are obtained from a staggered difference-in-differences specification following Callaway and Sant'Anna (2021). Standard errors, reported in parentheses, are clustered at the commuting zone level using multiplier bootstrap procedures with 999 replications. Statistical significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

2 Additional Enrollment Evidence

Figure A3: Event-Study Estimates: Effects of Amazon's Entry on Enrollment in 4-Year Public Colleges by Race



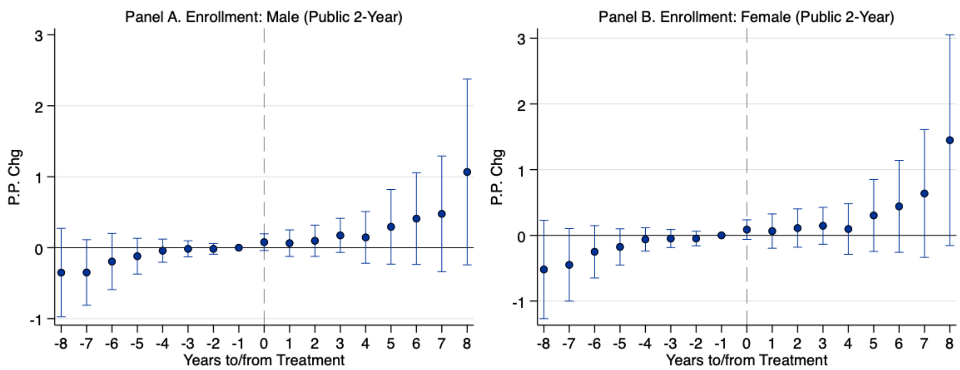
Notes: This figure presents dynamic estimates of Amazon fulfillment center entry on enrollment rates in four-year public colleges, disaggregated by race (White, Black, Hispanic). Enrollment data are from IPEDS (2010–2023); population denominators from ACS 1-Year estimates. Estimates follow Callaway and Sant'Anna (2021), with 95% confidence intervals from multiplier bootstrap standard errors clustered at the commuting zone level (999 replications).

Table A2: Effects of Amazon's Entry on Enrollment in 4-Year Public Colleges by Race

	White Students	Black Students	Hispanic Students
Amazon Entry	0.367**	0.061	0.061
	(0.173)	(0.039)	(0.061)
Mean Dep. Var.	10.20	1.39	1.41
SD	14.01	2.46	3.14
N	2,268	2,268	2,268

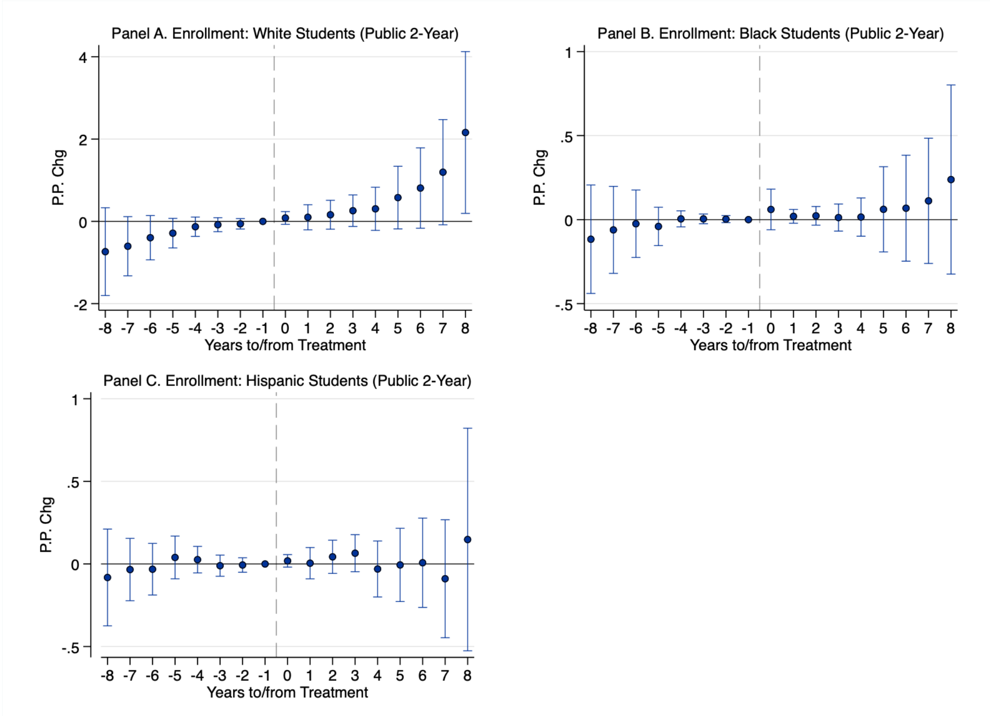
Notes: Average treatment effects on enrollment rates in four-year public colleges, disaggregated by race. Data as in Table A1. Standard errors clustered at the commuting zone level (multiplier bootstrap, 999 replications). Statistical significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure A4: Event-Study Estimates: Effects of Amazon's Entry on Enrollment in 2-Year Public Colleges



Notes: Dynamic estimates of Amazon fulfillment center entry on enrollment rates in two-year public colleges. Enrollment data from IPEDS (2010–2023); population denominators from ACS. Estimates follow Callaway and Sant'Anna (2021).

Figure A5: Event-Study Estimates: Effects of Amazon's Entry on Enrollment in 2-Year Public Colleges by Race



Notes: Dynamic estimates of Amazon fulfillment center entry on two-year public college enrollment rates, disaggregated by race. Data and estimator as in Figure A4.

Table A3: Effects of Amazon's Entry on Enrollment in 2-Year Public Colleges

	Male	Female	White	Black	Hispanic
Amazon Entry	0.225	0.253	0.415	0.054	0.025
	(0.169)	(0.185)	(0.284)	(0.056)	(0.069)
Mean Dep. Var.	2.92	3.91	4.22	0.98	0.83
SD	2.17	2.79	3.62	1.24	1.34
N	2,268	2,268	2,268	2,268	2,268

Notes: Average treatment effects on two-year public college enrollment rates, disaggregated by gender and race. Standard errors clustered at the commuting zone level (multiplier bootstrap, 999 replications). Statistical significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

3 Robustness Checks

Table A4: Robustness Check: Effects of Amazon's Entry on Educational Attainment (County-Level)

	High School+	Some College+	Bachelor's+	Master's+
Amazon Entry	0.155	0.333	0.366**	0.216**
	(0.137)	(0.211)	(0.145)	(0.109)
Mean Dep. Var.	88.80	61.57	31.62	11.94
SD	4.78	8.05	8.96	4.58
N	3,982	3,982	3,982	3,982

Notes: Re-estimated using county-level variation instead of commuting zones. The estimation sample consists of a balanced panel of 343 counties that eventually receive Amazon fulfillment centers. Estimates follow Callaway and Sant'Anna (2021), comparing treated counties to not-yet-treated counties with county and year fixed effects. The smaller magnitudes relative to commuting-zone estimates are consistent with positive spillovers across county boundaries within regional labor markets.

4 Balance Tests

Table A5: Balance Tests for Educational Outcomes in First-Differences by Treatment Group

Outcome (First-Differences)	2010–2013 (Cohort 1) Treat	Control	P-Value	2014–2016 (Cohort 2) Treat	Control	P-Value	2017–2019 (Cohort 3) Treat	Control	P-Value
Panel A. Less than High School									
Combined	-0.3854	-0.2883	0.154	-0.2817	-0.1654	0.064	-0.3805	-0.2976	0.296
Men	-0.3973	-0.2620	0.083	-0.2506	-0.0926	0.080	-0.3966	-0.3651	0.760
Women	-0.3725	-0.3122	0.425	-0.3130	-0.2312	0.202	-0.3630	-0.2344	0.166
Panel B. High School Degree or More									
Combined	0.3854	0.2883	0.154	0.2817	0.1654	0.064	0.3805	0.2976	0.296
Men	0.3973	0.2620	0.083	0.2506	0.0926	0.080	0.3966	0.3651	0.760
Women	0.3725	0.3122	0.425	0.3130	0.2312	0.202	0.3630	0.2344	0.166
Panel C. Some College or More									
Combined	0.5137	0.6322	0.233	0.4898	0.5009	0.918	0.4205	0.1940	0.109
Men	0.3974	0.5488	0.110	0.3569	0.3058	0.707	0.3104	0.0414	0.141
Women	0.6230	0.7110	0.497	0.6152	0.6798	0.579	0.5210	0.3428	0.263
Panel D. Bachelor's Degree or More									
Combined	0.3374	0.5950	0.000	0.4838	0.5436	0.433	0.5250	0.4907	0.755
Men	0.1713	0.4124	0.002	0.3203	0.3784	0.562	0.3697	0.3718	0.988
Women	0.4920	0.7642	0.004	0.6404	0.6942	0.520	0.6720	0.6001	0.612
Panel E. Master's Degree or More									
Combined	0.1718	0.3116	0.001	0.2250	0.2504	0.678	0.2233	0.2608	0.653
Men	0.0777	0.1986	0.025	0.1465	0.1913	0.539	0.1517	0.0993	0.596
Women	0.2596	0.4163	0.003	0.3004	0.3033	0.966	0.2919	0.4121	0.275

Notes: Balance tests for pre-treatment trends in educational outcomes across treatment cohorts, defined by year of first Amazon FC entry. Following Marcus and Sant'Anna (2021), these tests directly assess whether pre-treatment trends are balanced between treatment and control groups.

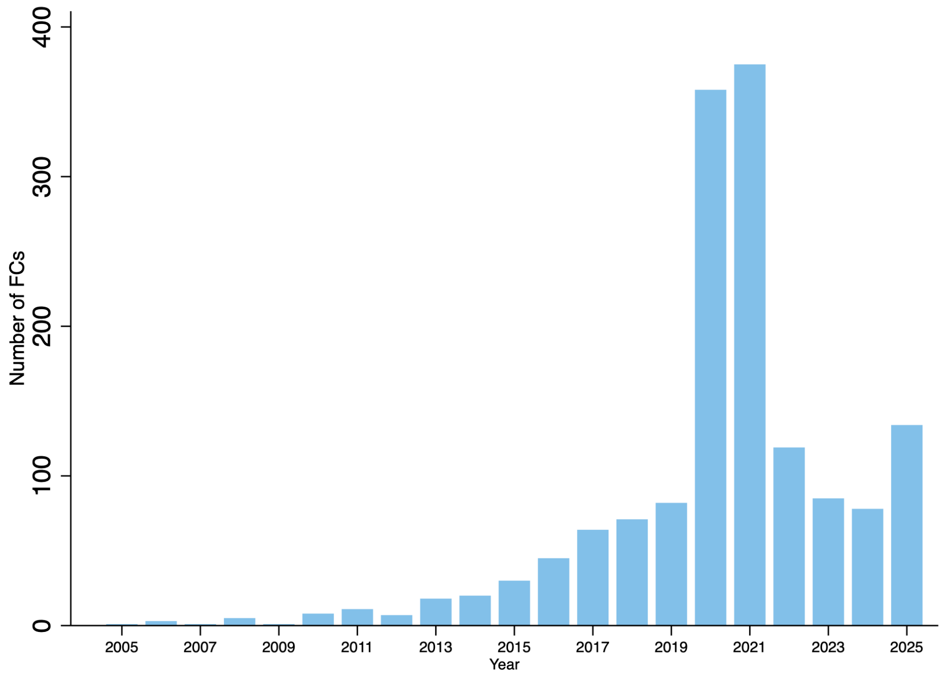
Table A6: Balance Tests for Demographic Variables in First-Differences by Treatment Group

	2010–2013 (Cohort 1)			2014–2016 (Cohort 2)			2017–2019 (Cohort 3)		
Variable (First-Differences)	Treat	Control	P-Value	Treat	Control	P-Value	Treat	Control	P-Value
Total Population	6380.69	20278.86	0.002	5996.46	7384.71	0.454	5531.12	1824.94	0.009
Male Population	3278.07	10575.26	0.002	2902.92	3666.60	0.395	2742.14	884.99	0.009
Female Population	3102.62	9703.60	0.003	3093.54	3718.11	0.519	2788.98	939.95	0.009
Male Population, Age 18-24	342.34	682.43	0.242	-186.47	-218.93	0.806	-226.81	-110.27	0.426
Male Population, Age 25-34	822.16	2551.91	0.012	719.82	937.79	0.348	640.68	228.23	0.012
Female Population, Age 18-24	273.19	507.62	0.405	-211.35	-193.33	0.884	-133.91	-130.52	0.977
Female Population, Age 25-34	631.89	2167.00	0.006	698.76	800.41	0.623	434.49	118.49	0.016
White Population	3244.67	9651.30	0.234	2513.02	5146.75	0.142	1956.19	-406.65	0.146
Black Population	727.13	3043.00	0.019	1004.76	1189.92	0.655	752.23	370.62	0.301
Hispanic Population	2596.14	6714.77	0.040	2991.57	3110.35	0.917	2894.73	768.49	0.012
Female-to-Male Labor Force Ratio	-0.85	-0.21	0.836	-4.29	-2.15	0.424	-3.58	-4.11	0.816
Labor Force (16+)	4164.64	11581.08	0.017	3118.09	3613.54	0.626	4626.00	1389.14	0.002

Notes: Balance tests for pre-treatment trends in demographic characteristics across treatment cohorts. Following Marcus and Sant'Anna (2021). While some demographic variables show statistically significant differences in certain cohorts, reflecting heterogeneity in the size and composition of treated versus control commuting zones, the consistent patterns across cohorts support the validity of the identification strategy.

5 Additional Descriptive Evidence

Figure A6: Amazon Fulfillment Center Openings by Year



Notes: Annual number of Amazon fulfillment center and warehouse openings in the United States, 2005–2025. Facility openings during 2020–2021 account for approximately 43% of total network expansion in the sample period.

6 Technical Details: Callaway and Sant'Anna Difference-in-Differences Estimator

This appendix provides technical details on the Callaway and Sant'Anna (2021) difference-in-differences estimator used throughout the analysis. The CS-DiD approach is specifically designed for staggered adoption settings where units are treated at different points in time and offers several advantages over traditional two-way fixed effects (TWFE) estimators.

6.1 Motivation and Key Advantages

Recent econometric research has demonstrated that standard TWFE estimators can produce misleading estimates in staggered adoption settings when treatment effects are heterogeneous across units or over time (Goodman-Bacon, 2021; Sun and Abraham, 2021). The fundamental issue arises because TWFE estimators implicitly use already-treated units as controls for newly-treated units. This creates "negative weights" on some group-time average treatment effects, potentially biasing the overall estimate even when all individual treatment effects are positive.

The Callaway and Sant'Anna estimator addresses these concerns through three key features. First, it constructs treatment effect estimates by comparing each newly-treated cohort exclusively to not-yet-treated units, avoiding contamination from already-treated observations. Second, it provides transparent aggregation of group-time specific effects into interpretable summary parameters while explicitly accounting for treatment timing variation. Third, it enables valid inference through a multiplier bootstrap procedure that properly accounts for clustering and estimation uncertainty.

6.2 Setup and Notation

Consider a balanced panel dataset with units $i = 1, \dots, N$ observed over time periods $t = 1, \dots, T$. Each unit i is associated with a commuting zone c, and I cluster standard errors at the commuting zone level. Let $G_c \in \{2, \dots, T, \infty\}$ denote the time period when commuting zone c first receives treatment (an Amazon fulfillment center), where $G_c = \infty$ indicates the commuting zone is never treated during the sample period.

Define D_{ct} as a binary treatment indicator:

$$D_{ct} = \mathbf{1}\{G_c \leq t\} \tag{A1}$$

which equals one if commuting zone c has been treated by period t, and zero otherwise.

For each unit, define potential outcomes $Y_{it}(0)$ and $Y_{it}(g)$ for $g \in \{2, \dots, T\}$, representing the outcome that would be observed in period t if the unit were never treated or first treated in period g, respectively. The observed outcome is:

$$Y_{it} = D_{ct} \cdot Y_{it}(G_c) + (1 - D_{ct}) \cdot Y_{it}(0) \tag{A2}$$

Following the main text, I account for anticipation effects by defining treatment as occurring one year prior to the first operational date of the fulfillment center. This means G_c is set to the calendar year before the facility opens.

6.3 Group-Time Average Treatment Effects

The fundamental building block of the CS-DiD framework is the group-time average treatment effect on the treated (ATT), denoted $ATT(g,t)$. This parameter measures the average treatment effect for the cohort of commuting zones first treated in period g, evaluated in calendar period t:

$$ATT(g,t) = \mathbb{E}[Y_t(g) - Y_t(0)|G_c = g] \tag{A3}$$

The parameter $ATT(g,t)$ captures the average difference between the observed outcome and the counterfactual untreated outcome for commuting zones in treatment cohort g during period t. When $t \geq g$, this represents a post-treatment effect; when $t < g$, it serves as a pre-treatment placebo test.

6.4 Identification

The key identifying assumption is parallel trends, which in the CS-DiD framework is stated in terms of changes in untreated potential outcomes:

$$\mathbb{E}[Y_t(0) - Y_{t-1}(0)|G_c = g] = \mathbb{E}[Y_t(0) - Y_{t-1}(0)|G_c > t] \tag{A4}$$

This assumption requires that, in the absence of treatment, the expected change in outcomes for cohort g would have been the same as the expected change for commuting zones not yet treated by period t. Importantly, this comparison uses only not-yet-treated units as controls, avoiding the contamination from already-treated units that plagues TWFE estimators.

Under this parallel trends assumption, combined with standard regularity conditions, the group-time ATT is point-identified. The CS-DiD estimator implements this identification strategy using a doubly-robust approach that combines outcome regression and inverse probability weighting.

6.5 Estimation

For each (g,t) pair with $t \geq g$, the doubly-robust estimator of $ATT(g,t)$ is:

$$\begin{aligned}\widehat{ATT}^{DR}(g,t) = \frac{1}{N_g} \sum_{i:G_{c(i)}=g} \Bigg[& \left(Y_{it} - Y_{i,g-1}\right) \\ & - \left(1 - \frac{\widehat{p}_{c(i)}}{\widehat{p}_{c(i)}}\right) \left(Y_{it} - Y_{i,g-1} - \widehat{m}(t,g-1,X_i)\right) \Bigg] \\ & - \frac{1}{N_{\text{comp}}} \sum_{i:G_{c(i)}>t} \frac{\widehat{p}_{c(i)}}{1-\widehat{p}_{c(i)}} \left(Y_{it} - Y_{i,g-1} - \widehat{m}(t,g-1,X_i)\right)\end{aligned} \tag{A5}$$

where:

- N_g is the number of units in cohort g
- N_{comp} is the number of comparison units (not yet treated by period t)
- $\widehat{p}_{c(i)} = \widehat{P}(G_{c(i)} = g | G_{c(i)} \in \{g, \text{not yet treated}\}, X_i)$ is the estimated propensity score
- $\widehat{m}(t, g-1, X_i) = \widehat{\mathbb{E}}[Y_{it} - Y_{i,g-1} | G_{c(i)} > t, X_i]$ is the estimated outcome regression

The doubly-robust estimator has the property that it remains consistent if either the propensity score model or the outcome regression model is correctly specified. In my baseline implementation, I do not include covariates (X_i), setting the propensity score to the sample proportion and omitting the outcome regression adjustment. This choice reflects the view that most local economic and demographic variables are potentially affected by Amazon's entry, and including them would risk post-treatment bias.

For pre-treatment periods ($t < g$), I estimate placebo effects using:

$$\widehat{ATT}(g,t) = \mathbb{E}[Y_t - Y_{g-2} | G_c = g] - \mathbb{E}[Y_t - Y_{g-2} | G_c > t+1] \tag{A6}$$

These pre-treatment estimates provide visual evidence on the plausibility of parallel trends and are displayed in all event-study figures.

6.6 Aggregation

The group-time ATTs are aggregated into summary parameters for presentation and interpretation. I report two primary aggregations.

6.6.1 Event-Study Parameters

To examine dynamic treatment effects, I aggregate group-time ATTs by event time $e = t - g$, which measures periods relative to treatment:

$$\theta_{ES}(e) = \sum_{g \in \mathcal{G}} \mathbf{1}\{g + e \leq T\} \cdot \frac{N_g}{\sum_{g' \in \mathcal{G}} \mathbf{1}\{g' + e \leq T\} N_{g'}} \cdot ATT(g, g + e) \tag{A7}$$

where $\mathcal{G}$ is the set of treatment cohorts and the weights are proportional to cohort sizes. This aggregation produces the event-study plots presented throughout the paper, with $e = -1$ normalized to zero as the reference period.

6.6.2 Overall Average Treatment Effect

The simple weighted average ATT aggregates all post-treatment group-time effects:

$$\theta_{ATT} = \sum_{g \in \mathcal{G}} \sum_{t \geq g} \mathbf{1}\{t \leq T\} \cdot w(g, t) \cdot ATT(g, t) \tag{A8}$$

where the weights $w(g, t)$ are proportional to the size of cohort g and the number of post-treatment periods. This is the primary parameter reported in regression tables and represents the average effect across all treated commuting zones and post-treatment periods.

References

Callaway, B. and P. H. Sant'Anna (2021): "Difference-in-Differences with Multiple Time Periods," *Journal of Econometrics*, 225, 200–230.

Goodman-Bacon, A. (2021): "Difference-in-Differences with Variation in Treatment Timing," *Journal of Econometrics*, 225, 254–277.

Marcus, M. and P. H. C. Sant'Anna (2021): "The Role of Parallel Trends in Event Study Settings: An Application to Environmental Economics," *Journal of the Association of Environmental and Resource Economists*, 8, 235–275.

Sun, L. and S. Abraham (2021): "Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects," *Journal of Econometrics*, 225, 175–199.